

Assignment Day 3

Q)1. Implement the following using i) 2:1 Mux and NOT gates ii) 4:1 Mux

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

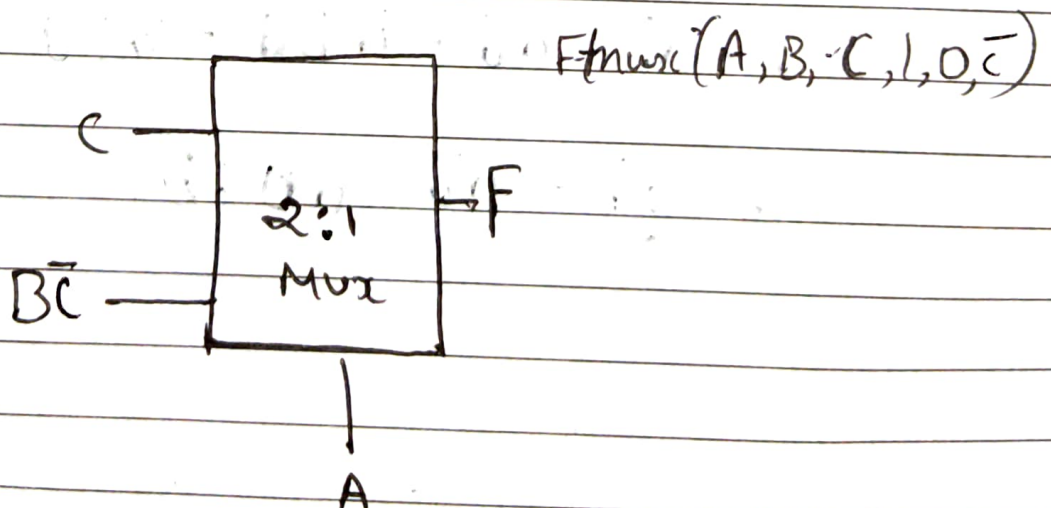
Ans: i) when $A=0$

$$F = C + B\bar{C} = C + B$$

when $A=1$

$$F = B\bar{C}$$

select = A, $I_0 = C$, $I_1 = B\bar{C}$

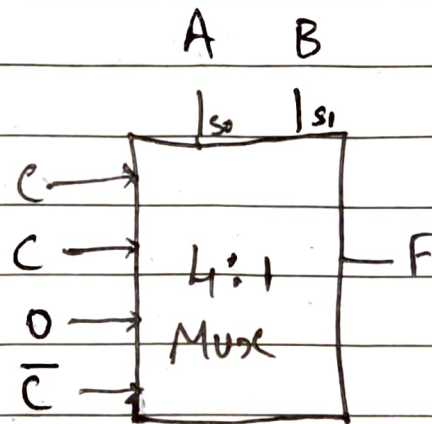


ii)

$$S_1 = A$$

$$S_0 = B$$

	A	B	F
$I_0 = 0$	0	0	C
$I_1 = 0$	0	1	C
$I_2 = 1$	1	0	0
$I_3 = 1$	1	1	$\bar{C}$



$$b) F(P, Q, R) = \bar{P}\bar{R} + QR + \bar{P}Q$$

Ans:

i)

$$P = 0$$

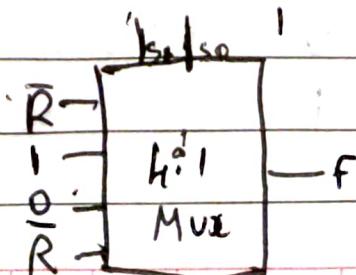
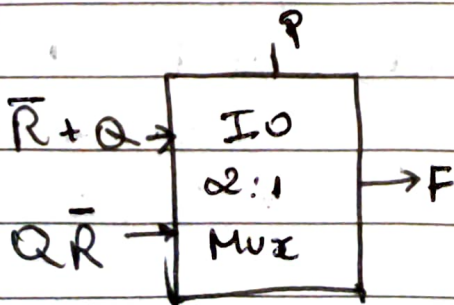
$$P = 1$$

$$F = \bar{R} + Q$$

$$F = QR$$

ii)

P	Q	S
0	0	$\bar{R}$
0	1	1
1	0	0
1	1	$\bar{R}$





Q 2) Implement the functionality of a full adder using 2:1 mux only.

~~Ans:~~ $S = A \oplus B \oplus C_{in}$

$$C_{out} = AB + BC_{in} + AC_{in}$$

$$\text{select} = A$$

$$I_0 = B$$

$$I_1 = \bar{B}$$

$$X = A \oplus B$$

$$S = (A \oplus B) \oplus C_{in}$$

$$X = A \oplus B$$

$$S = X \oplus C_{in}$$

Using 2:1 mux

$x \rightarrow$ select line

$$S = \bar{x}(C_{in}) + x(\bar{C}_{in})$$

$$S = x$$

$$I_0 = C_{in} \quad I_1 = \bar{C}_{in}$$

$$S = (A \oplus B) \oplus C_{in}$$

Carry: $C_{out} = AB + AC_{in} + BC_{in}$

$$F = \bar{S}I_0 + SI_1$$

$$X = A \oplus B$$

$$x = 0$$

when $A=B=0$

$$X = 1 \quad \text{when } A=0, B=1$$

$$A = B$$

$$C_{out} = 0$$

$$A \neq B \quad C_{out} = 0 + 0 + C_{in} +$$

$$A \quad B$$

When $A=B=1$, $C_{out}=1$

$$A \cdot C_{in} = C_{in}$$

$$0 \quad 0$$

whenever $X=0$

$$0 \quad 1$$

$$A=1, B=0$$

$$\bar{0} \quad 1$$

$$C_{out} = A$$

$$1 \quad 0$$

$$C_{out} = 0 + 1 + C_{in} + 0$$

$$= C_{in}$$

$$C_{out} = C_{in}$$

$$y = \bar{5}I_0 + 5I_1$$

$$x = A \oplus B$$

$$x = 0 \Rightarrow A$$

$$x = 1 \Rightarrow C_{in}$$

$$C_{out} = \bar{x}A + C_{in}$$

$$I_0 = A \quad C_{out} = (A \oplus B)A +$$

$$I_1 = C_{in} \quad \underline{(A \oplus B) \cdot C_{in}}$$

Q.3) Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$ and implement the SOP expression using 4:1 mux.

cd \ ab	00		01		10		11	
	00	01	10	11	00	01	10	11
00							1	
01	1		1				1	
11	x		x		x		x	
10							1	

$$\underline{\underline{b\bar{c} + cd}}$$

Implement using 4:1 mux
choose B & C as select line.

$$S_1 = B$$

$$S_0 = C$$

$$BC = 00, \text{ then } F = 0 \Rightarrow I_0 = 0$$

$$BC = 01, \text{ then } F = D \Rightarrow I_1 = D$$

$$BC = 10, \text{ then } F = 1 \Rightarrow I_2 = 1$$

$$BC = 11, \text{ then } F = \overline{D} \Rightarrow I_3 = \overline{D}$$

$$F = \underline{B\overline{C} + C\overline{D}}$$